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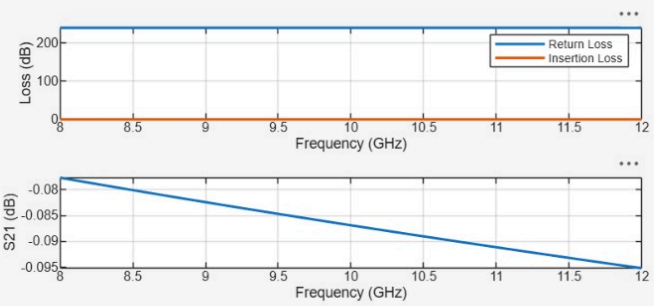
m1.m | m2.m

/MATLAB Drive/m2.m

```
clc; clear; close all;
f = (8:0.1:12)*1e9; f0 = 10e9; Z0 = 50; L = 0.05;
c = 3e8; vp = c/sqrt(2.1); alpha = 0.2*sqrt(f/f0);
beta = 2*pi*f/vp; gamma = alpha + 1j*beta;
S11 = zeros(size(f)); S22 = S11;
S21 = exp(-gamma*L); S12 = S21;
RL = -20*log10(max(abs(S11),1e-12));
IL = -20*log10(abs(S21)); eta = abs(S21).^2*100;
subplot(2,1,1); plot(f/1e9,RL,f/1e9,IL,'LineWidth',1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('Loss (dB)');
legend('Return Loss','Insertion Loss');
subplot(2,1,2); plot(f/1e9,20*log10(abs(S21)),'LineWidth',1.5);
grid on; xlabel('Frequency (GHz)'); ylabel('S21 (dB)');
[~,k] = max(abs(S21)); [~,m] = min(abs(S11));
```

Apps

Figure 1



Command Window

```
Best transmission frequency = 8.0 GHz
Minimum reflection occurs throughout sweep (ideal matching).
>>
```

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